

$$\begin{aligned}
 & x(t) \rightarrow \boxed{h(t)} \rightarrow y(t) \quad \text{circled: } x(t-t_0) * h(t) = h(t-t_0) \\
 & \quad \quad \quad \text{circled: } x(t) * h(t) = h(t) \\
 & \quad \quad \quad \text{circled: } \Delta = \Delta \\
 & s(t) \rightarrow \boxed{h(t)} \rightarrow h(t) \\
 & \textcircled{1} \text{ Linear} \\
 & \textcircled{2} \text{ Time-invariant} \\
 & \text{LTI System} \\
 & x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau \\
 & \quad \quad \quad \uparrow \quad \quad \quad \uparrow \text{freeze} \\
 & \quad \quad \quad \text{input} \quad \quad \text{system} \\
 & \quad \quad \quad \text{int} \quad \quad \text{filter}
 \end{aligned}$$

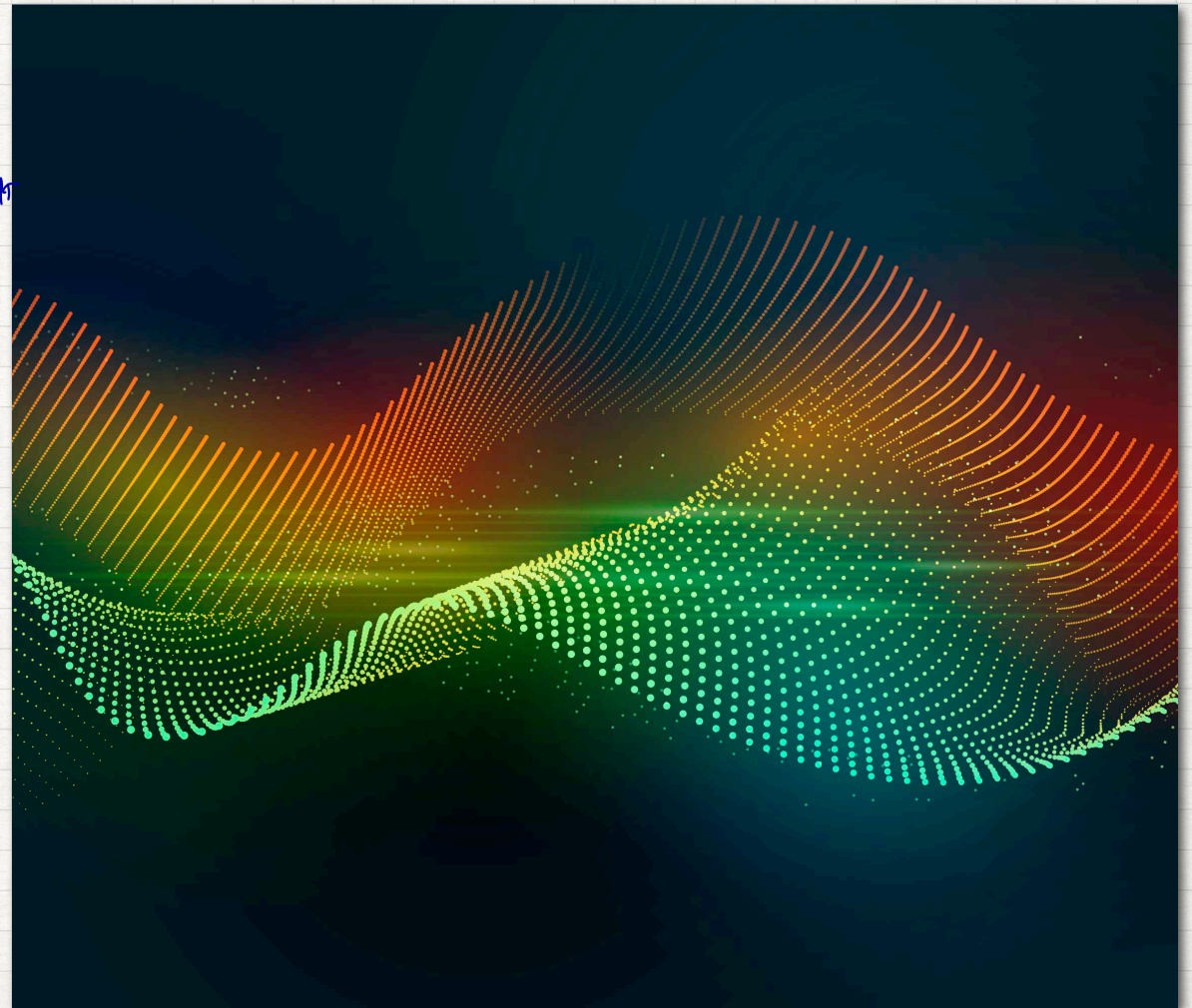
LECTURE 03

FOURIER

ANALYSIS OF

CONTINUOUS-

TIME SIGNALS



INTRODUCTION



Time



Frequency

- **Continuous-time Signals (t)**

- Periodic signals - $x(t)$

- Aperiodic signals - $x(t)$

$x(t)$

a_n

$x(t)$

$X(j\omega)$

- **Discrete-time Signals [n]**

- Periodic signals - $x[n]$

- Aperiodic signals - $x[n]$

- DFT - $x[n]$

$x[n]$

a_n

$x[n]$

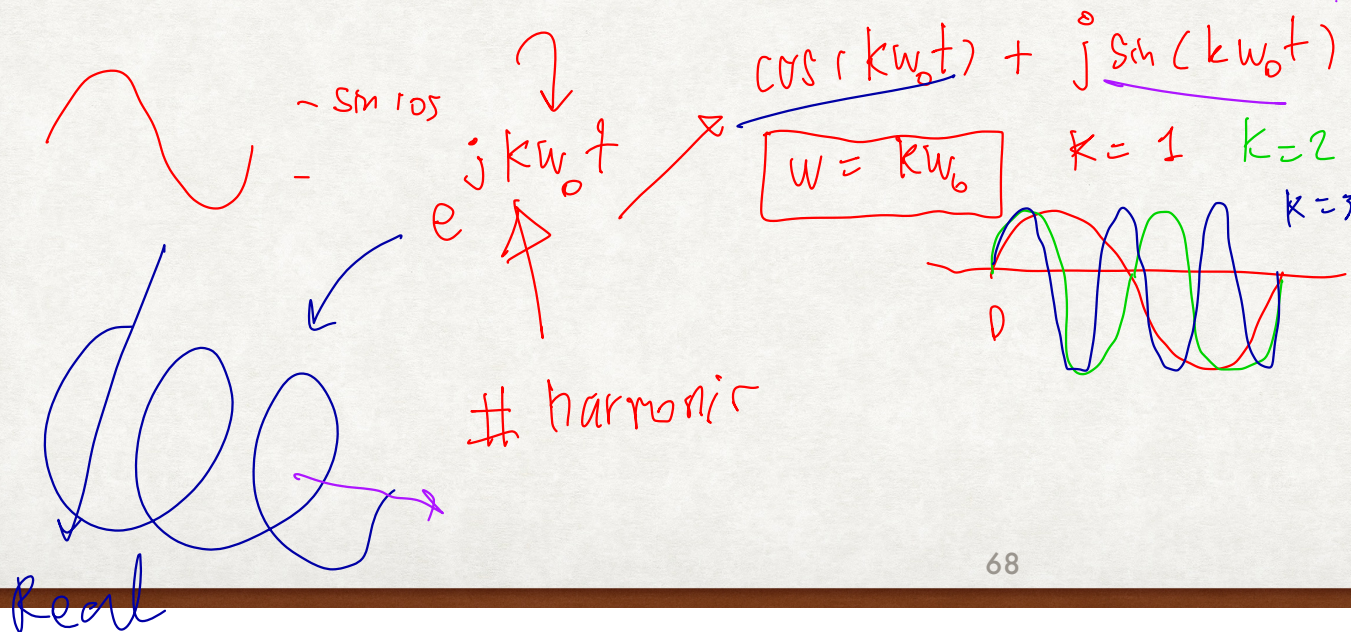
$X(e^{j\omega})$

$x[n]$

$X[k]$

OUTLINE

- (1) • Fourier series representation of continuous-time periodic signals
- (2) • Continuous-time Fourier transform
- (3) • Properties of Continuous-time Fourier transform



$k_{w,j}$
 $e^{j\omega_0 t}$
 $e^{(a+bj)t}$
 constant
 $a + bj$

Response of LTI systems to complex exponentials.

Complex exponential signals e^{st} where s are complex numbers

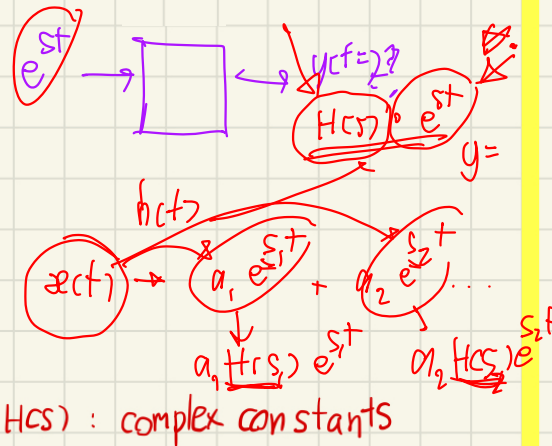
Let $x(t) = e^{st}$

$$y(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau$$

$$= e^{st} \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$$

$H(s)$



$$y(t) = e^{st} H(s)$$

$H(s) \rightarrow$ eigenvalue
 $e^{st} \rightarrow$ eigenfunction

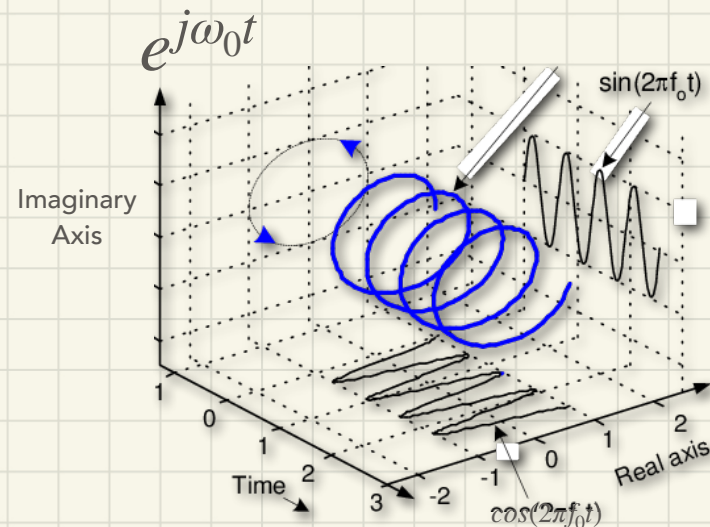
Response of complex expo is the same complex expo with only a change in amplitude.

In a linear system, a signal can be decomposed in terms of eigenfunctions. Let $x(t)$ correspond to a linear combination of three complex exponentials:

$$x(t) = a_1 e^{s_1 t} + a_2 e^{s_2 t} + a_3 e^{s_3 t}$$

$a_1 H(s_1) e^{s_1 t}$ $a_2 H(s_2) e^{s_2 t}$ $a_3 H(s_3) e^{s_3 t}$

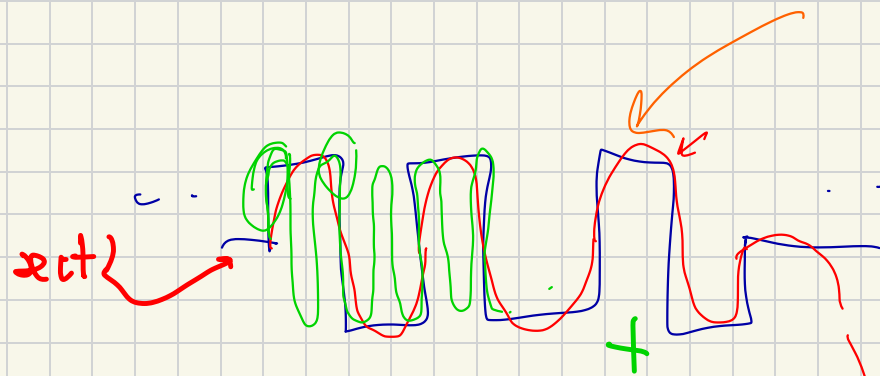
$$y(t) = a_1 H(s_1) e^{s_1 t} + a_2 H(s_2) e^{s_2 t} + a_3 H(s_3) e^{s_3 t}$$



FOURIER SERIES REPRESENTATION

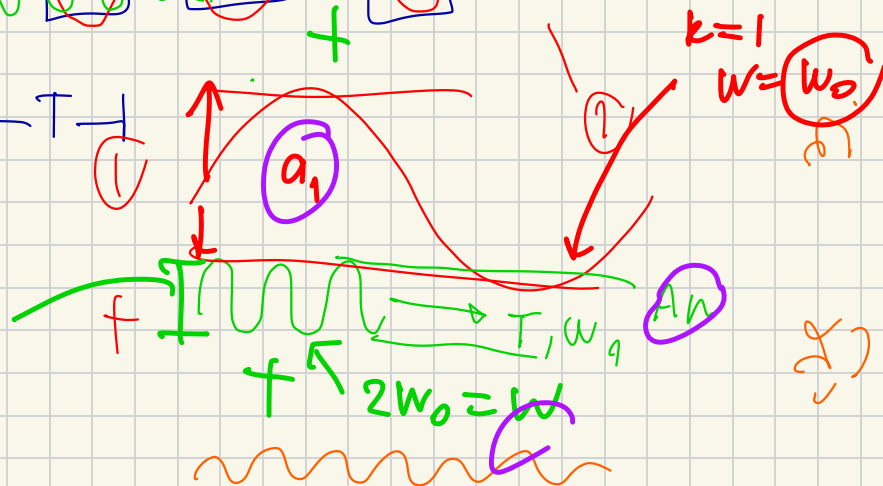


- Analyse periodic signals and representing a signal over a finite interval in terms of some set of orthogonal basis.
- "Fourier" named after French physicist, "Jean Baptiste Joseph Fourier" (1768-1830), suggested
 - " Periodic signal could be represented by a sum of sinusoids"
- Learn to
 - Decompose periodic signals into their frequency components, for application examples, acoustic and electromagnetic waves, vertical displacement of a mechanical pendulum, periodic vibrations of musical instruments, beautiful pattern of crystal structures.



$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

a_k



∞

FOURIER SERIES REPRESENTATION

- A signal is periodic if, for some positive non-zero value of T ,

$$x(t) = x(t + nT), \quad n = 1, 2, \dots \quad (3.1)$$

T is the fundamental period, $2\pi/T$ is fundamental radian frequency (ω_0)

- Sinusoidal signal $x(t) = \cos(\omega_0 t)$ and the periodic complex expo $x(t) = e^{j\omega_0 t}$ are periodic with fundamental frequency ω_0 , and the fundamental period $T = 2\pi/\omega_0$

- The equation below is the set of harmonically related complex exponentials

$$\phi_k(t) = e^{jk\omega_0 t} = e^{jk(2\pi/T)t}, \quad k = 0, \pm 1, \pm 2, \dots \quad (3.2)$$

real

- RIES REPRESENTATION

~~$a_0 e^{j(0)}$~~

$$n, k = 0,$$

- a_k is Fourier series coefficients. The term, $k = 0$, is a constant
- $k=1, -1 \rightarrow$ fundamental components / first harmonic components
- $k=2, -2 \rightarrow$ half period / twice frequency / second harmonic components
- ... $k=N, -N \rightarrow$ Nth harmonic components

Real $x^*(t) = x(t)$

$k = -1, +1$

$z = a + bj$
 $z^* = a - bj$

$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$

FOURIER SERIES REPRESENTATION

$z = ab \quad z^* = a^* b^*$

If $x(t)$ is real, then $x^*(t) = x(t)$ and we obtain

$x(t) = \sum_{k=-\infty}^{\infty} a_k^* e^{-jk\omega_0 t}$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t} \quad (3.3)$$

Replacing k with $-k$, we have

$x(t) = \sum_{k=-\infty}^{\infty} a_{-k}^* e^{jk\omega_0 t}$

$a_k = a_{-k}^*$
 $a_{-k} = a_k^*$

Compare above equation with (3.3), we get $a_k = a_{-k}^*$ or $a_k^* = a_{-k}$

To derive an alternative form of the Fourier series

$x(t) = a_0 + \sum_{k=1}^{\infty} [a_k e^{jk\omega_0 t} + a_{-k} e^{-jk\omega_0 t}]$

From (3.6) we get $x(t) = a_0 + \sum_{k=1}^{\infty} [a_k e^{jk\omega_0 t} + a_k^* e^{-jk\omega_0 t}]$

Since two terms inside the summation are complex conjugates of each other, this can be expressed as

$x(t) = a_0 + \sum_{k=1}^{\infty} 2\text{Re}\{a_k e^{jk\omega_0 t}\}$

Another form is obtained by writing a_k as

$a_k = B_k + jC_k$

where B_k and C_k are both real. With this expression, $x(t)$ takes the form

$x(t) = a_0 + 2 \sum_{k=1}^{\infty} [B_k \cos(k\omega_0 t) - C_k \sin(k\omega_0 t)]$

$e^{j\theta} = \cos\theta + j\sin\theta$

$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$

$$\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

HOW TO FIND a_k ?

FOURIER SERIES REPRESENTATION

Assuming that a given periodic signal can be represented the series of eq. (3.3)

Now we have $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$ (3.3)

Multiply both by $e^{-jn\omega_0 t}$ we obtain

$$x(t)e^{-jn\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} e^{-jn\omega_0 t} \quad (3.4)$$

Integrate both side from 0 to $T = 2\pi/\omega_0$

$$\int_0^T x(t)e^{-jn\omega_0 t} dt = \int_0^T \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} e^{-jn\omega_0 t} dt \quad (3.5)$$

Interchanging order of integration and summation yields

$$\int_0^T x(t)e^{-jn\omega_0 t} dt = \sum_{k=-\infty}^{\infty} a_k \left[\int_0^T e^{jk\omega_0 t} e^{-jn\omega_0 t} dt \right] \quad (3.6)$$

From Euler's relation, we obtain

$$\int_0^T e^{j(k-n)\omega_0 t} dt = \frac{e^{j(k-n)\omega_0 t}}{j(k-n)\omega_0} \Big|_{t=0}^T \quad (3.7)$$

For $k \neq n$ the integral on the right-hand side of (3.6) are zero, for $k = n$, the integral equals T .

$$\int_0^T e^{j(k-n)\omega_0 t} dt = \begin{cases} T, & k = n \\ 0, & k \neq n \end{cases} \quad (3.8)$$

When $k = n$, (3.6) Reduces to $\int_0^T x(t)e^{-jn\omega_0 t} dt = a_n T$ (3.9)

We have $a_n = \frac{1}{T} \int_{\langle T \rangle} x(t)e^{-jn\omega_0 t} dt$ (3.10)

$$\int_0^T e^{j(k-n)\omega_0 t} dt = ?$$

$$e^{j0} = 1$$

$$= \frac{e^{j(k-n)\omega_0 t}}{j(k-n)\omega_0} \Big|_0^T = \frac{e^{j(k-n)\omega_0 T} - e^{j(k-n)\omega_0 \cdot 0}}{j(k-n)\omega_0}$$

$k \neq n$

= 0

$$e^{j(k-n) \cdot 2\pi} = 1$$

$$= \frac{\cos(k-n) \cdot 2\pi + j \sin(k-n) \cdot 2\pi}{0}$$

$$k=n \int_0^T 1 dt = t \Big|_0^T = T$$

FS coefficient weight

$$\int_0^T x(t) e^{-jn\omega_0 t} dt = a_n \cdot T$$

$$a_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt$$

$n \equiv k$

$$x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t}$$

$$\phi_k(t) = e^{jk\omega_0 t}$$

basis ϕ_n

$k, = 0, 1, 2, \dots, \infty$

FOURIER SERIES REPRESENTATION



- Set of complex exponentials form a orthogonal set:

$$\phi_n = e^{jn(2\pi/T)t} \quad (3.11)$$

- Representing signal $x(t)$ in terms of sum of basis functions

$$x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} a_n e^{jn(2\pi/T)t} \quad (3.12)$$

Fourier coefficients

$$a_n = \frac{1}{T} \int_{\langle T \rangle} x(t) e^{-jn\omega_0 t} dt = \frac{1}{T} \int_0^T x(t) e^{-jn(2\pi/T)t} dt \quad (3.13)$$

Handwritten annotations: The first equation (3.12) is enclosed in a green box. The term a_n is circled in green, and the term $e^{jn\omega_0 t}$ is circled in green. A purple arrow points from the a_n term to the coefficient equation below. The word 'dis' is written in purple next to the summation. The second equation (3.13) is also enclosed in a green box. The term $\frac{1}{T}$ is circled in green, and the integration limits $\langle T \rangle$ are circled in green. The word 'cont' is written in purple next to the integration. A purple arrow points from the a_n term in the first equation to the coefficient equation below.

FOURIER SERIES REPRESENTATION

• Exponential Fourier series

• Each frequency is a multiple of $\omega_0/2\pi$

• Coefficients a_n define a complex-valued function of the discrete frequency $n\omega_0$

• DC component is equal to a_0 and is obtained by setting $n = 0$

• The plot of $|a_n|$ vs. $n\omega_0$ displays the amplitude of various frequency — magnitude spectrum

• The plot of the phase of $a_n == \angle a_n$ versus $n\omega_0$ is called phase spectrum of $x(t)$

• The plots of $|a_n|$ and phase of a_n is called "line spectra"

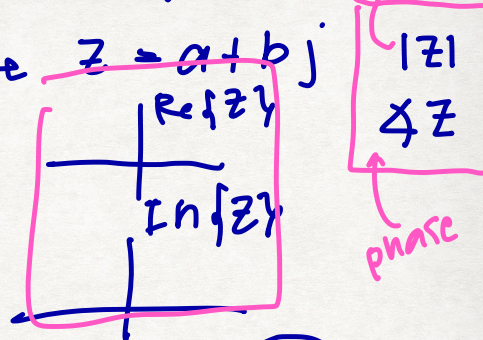
• For real-valued signals: $|a_n| = |a_{-n}|$ and $\angle a_n = -\angle a_{-n}$

$$a_k = a_{-k}^*$$

$$|a_k| = |a_{-k}| \quad \angle a_n = -\angle a_{-n}$$

$$\angle z = \tan^{-1}\left(\frac{b}{a}\right)$$

$$|z| = \sqrt{a^2 + b^2} \quad \text{mag}$$



$$x(t) \leftrightarrow a_n$$

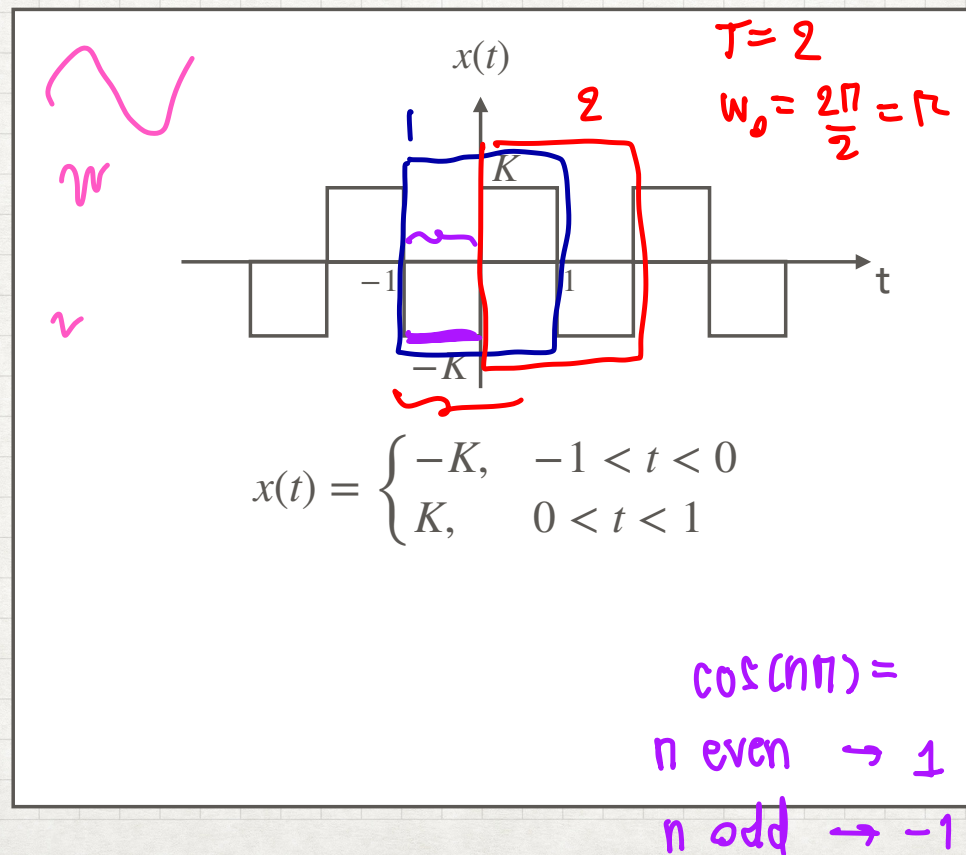
FOURIER SERIES REPRESENTATION

EXERCISE #3.1



a_n

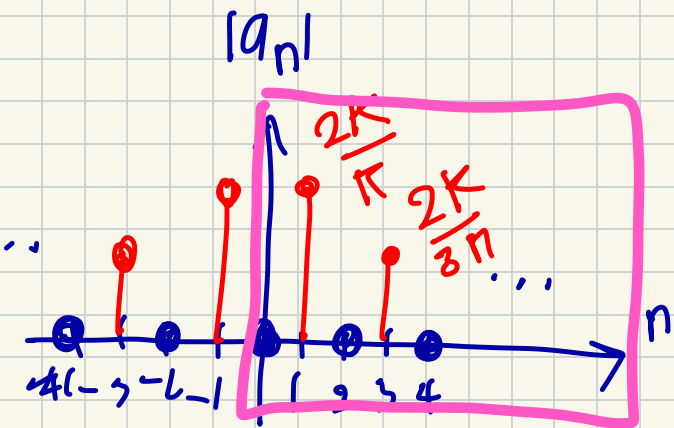
Find line spectra for the periodic signal $x(t)$



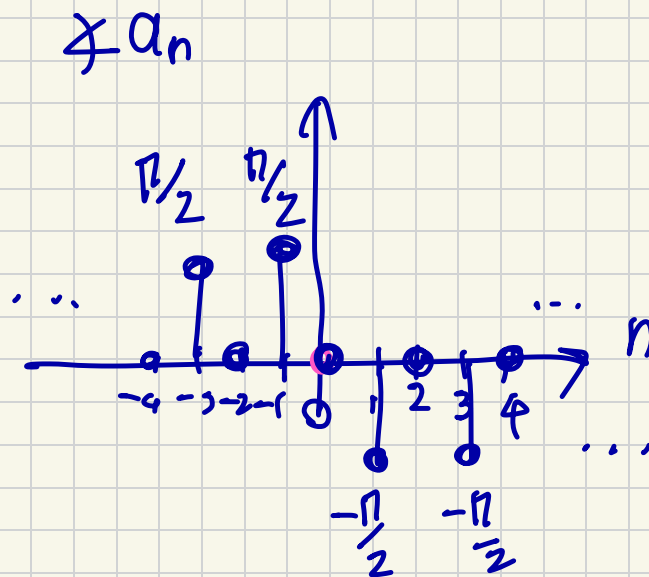
$$\begin{aligned}
 a_n &= \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt \\
 &= \frac{1}{2} \left(\int_{-1}^0 -K e^{-jn\pi t} dt + \int_0^1 K e^{-jn\pi t} dt \right) \\
 &= \frac{K}{2} \left(\left. \frac{e^{-jn\pi t}}{-jn\pi} \right|_{-1}^0 + \left. \frac{e^{-jn\pi t}}{-jn\pi} \right|_0^1 \right) \\
 &= \frac{K}{2jn\pi} \left(1 - e^{+jn\pi} - e^{-jn\pi} + 1 \right) \\
 &= \frac{K}{jn\pi} \left(1 - \frac{1}{2} (e^{jn\pi} + e^{-jn\pi}) \right) \\
 &= \frac{K}{jn\pi} (1 - \cos n\pi)
 \end{aligned}$$

$$\frac{1}{a^2 + b^2} \cdot \frac{(a-bj)(a+bj)}{(a-bj)(a+bj)}$$

$$= \begin{cases} 0 & , n \text{ even} \\ 2K/jn\pi & , n \text{ odd} \end{cases}$$



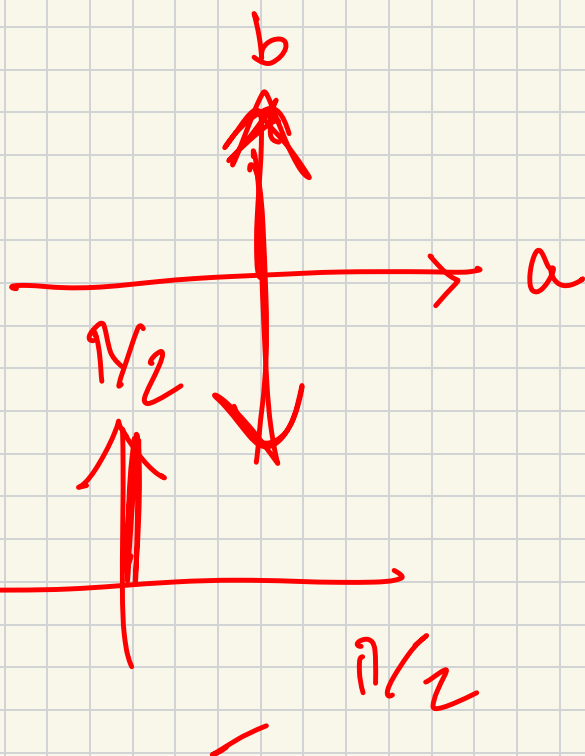
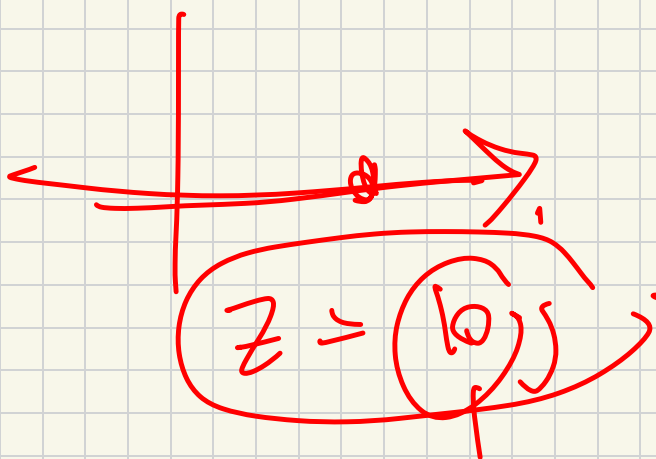
$$|a_n| = |a_{-n}|$$



$$a_n = -a_{-n}$$

$$z = a + bj$$

$$z = 10 \angle z$$



$$\left| \frac{2K}{jn\pi} \right|$$

$$\frac{2K}{jn\pi}$$

$$\frac{\pi}{2}$$

$$-\frac{\pi}{2}$$

$$\tan^{-1}\left(\frac{b}{a}\right)$$

$$n = +1$$

$$n = +1: \frac{2K}{jn\pi} = \frac{2K}{j\pi}$$

$$j = \sqrt{-1} \quad j^2 = -1$$

$$n = +3 \rightarrow \frac{-2K}{3n\pi}$$

$$b_j \tan^{-1}\left(\frac{-2K}{n\pi}\right)$$

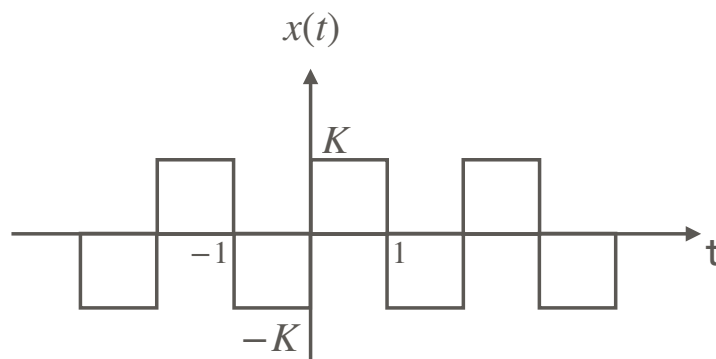
$$\ominus 2K/n \quad - \pi/2$$

FOURIER SERIES REPRESENTATION

EXERCISE #3.1



Find line spectra for the periodic signal $x(t)$

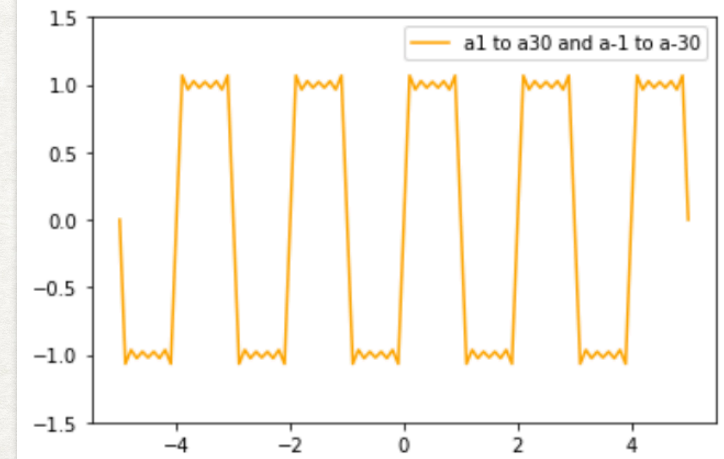
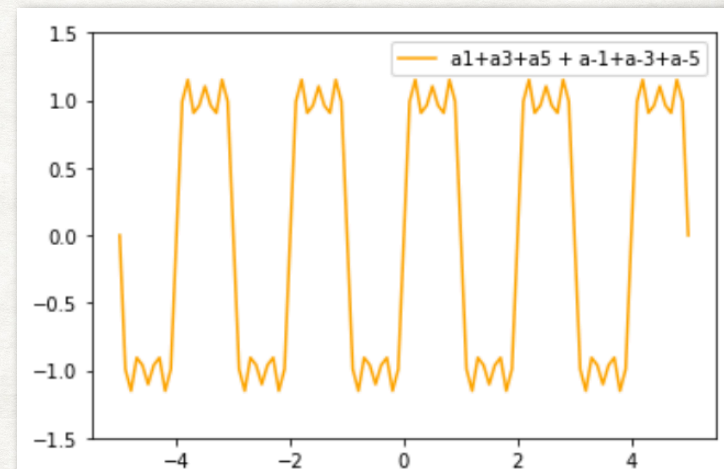
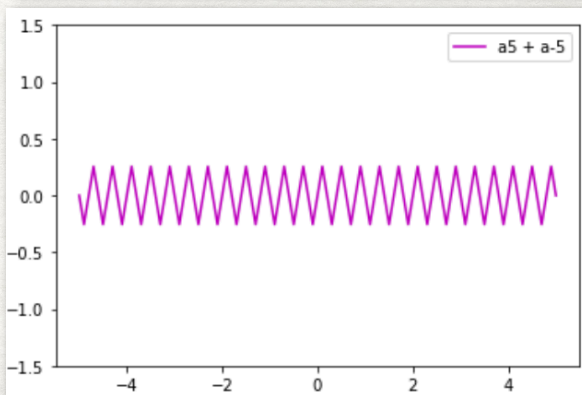
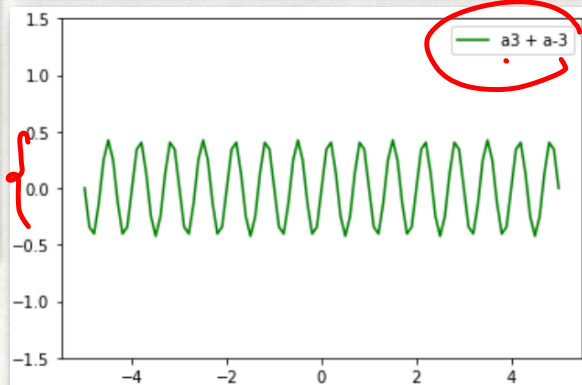
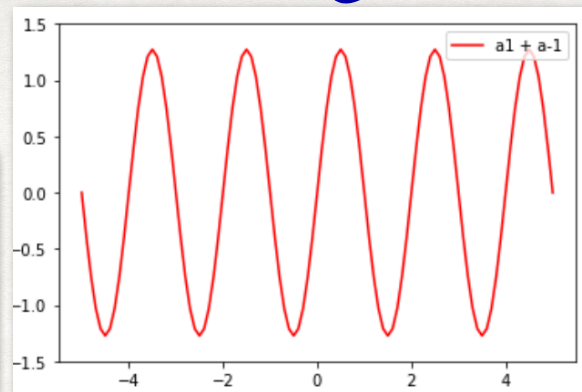
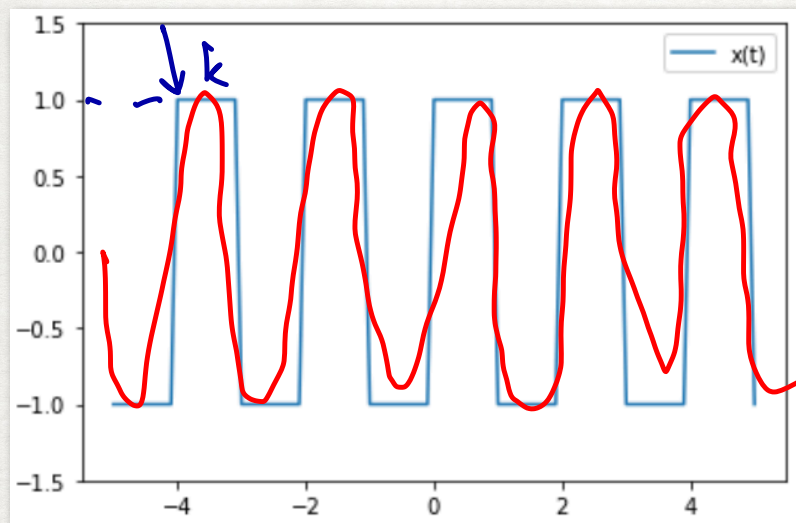


$$x(t) = \begin{cases} -K, & -1 < t < 0 \\ K, & 0 < t < 1 \end{cases}$$

and $x(t+2) = x(t)$. Therefore, $\omega_0 = 2\pi/2 = \pi$

$$\begin{aligned} \text{From } a_n &= \frac{1}{T} \int_{\langle T \rangle} x(t) e^{-jn(2\pi/T)t} dt, \\ &= \frac{1}{2} \int_{-1}^1 x(t) e^{-jn(2\pi/2)t} dt \\ &= \frac{1}{2} \left[\int_{-1}^0 -K e^{-jn\pi t} dt + \int_0^1 K e^{-jn\pi t} dt \right] \\ &= \frac{K}{jn\pi} \left[1 - \frac{1}{2}(e^{jn\pi} + e^{-jn\pi}) \right] \\ &= \begin{cases} \frac{2K}{jn\pi}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases} \end{aligned}$$

Pl



FOURIER SERIES REPRESENTATION

EXERCISE #3.2



Find line spectra for the periodic signal $x(t)$

$$a_n = \int \sin -$$

$$\sin =$$

$$\sin = \frac{e^{j\omega t} - e^{-j\omega t}}{2j}$$

$$x(t) = 1 + \sin(\omega_0 t) + 2\cos(\omega_0 t) + \cos(2\omega_0 t + \frac{\pi}{4})$$

e

$$a_n e^{jn\omega_0 t}$$

$$\sin \omega_0 t = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}$$

$$\cos \omega_0 t = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}$$

FOURIER SERIES REPRESENTATION

EXERCISE #3.2



Find line spectra for the periodic signal $x(t)$

1. $\omega_0 \rightarrow k=1$ 2. $2\omega_0 \rightarrow k=2$

$$x(t) = 1 + \sin(\omega_0 t) + 2\cos(\omega_0 t) + \cos(2\omega_0 t + \frac{\pi}{4})$$

$$= 1 + \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j} + \frac{2}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t}) + \frac{1}{2}(e^{j(2\omega_0 t + \frac{\pi}{4})} + e^{-j(2\omega_0 t + \frac{\pi}{4})})$$

$$= 1 - \frac{j}{2}e^{j\omega_0 t} + \frac{j}{2}e^{-j\omega_0 t} + e^{j\omega_0 t} + e^{-j\omega_0 t} + \frac{1}{2}e^{j\frac{\pi}{4}}e^{j2\omega_0 t} + \frac{1}{2}e^{-j\frac{\pi}{4}}e^{-j2\omega_0 t}$$

$$a_0 = 1; |a_0| = 1; \angle a_0 = 0$$

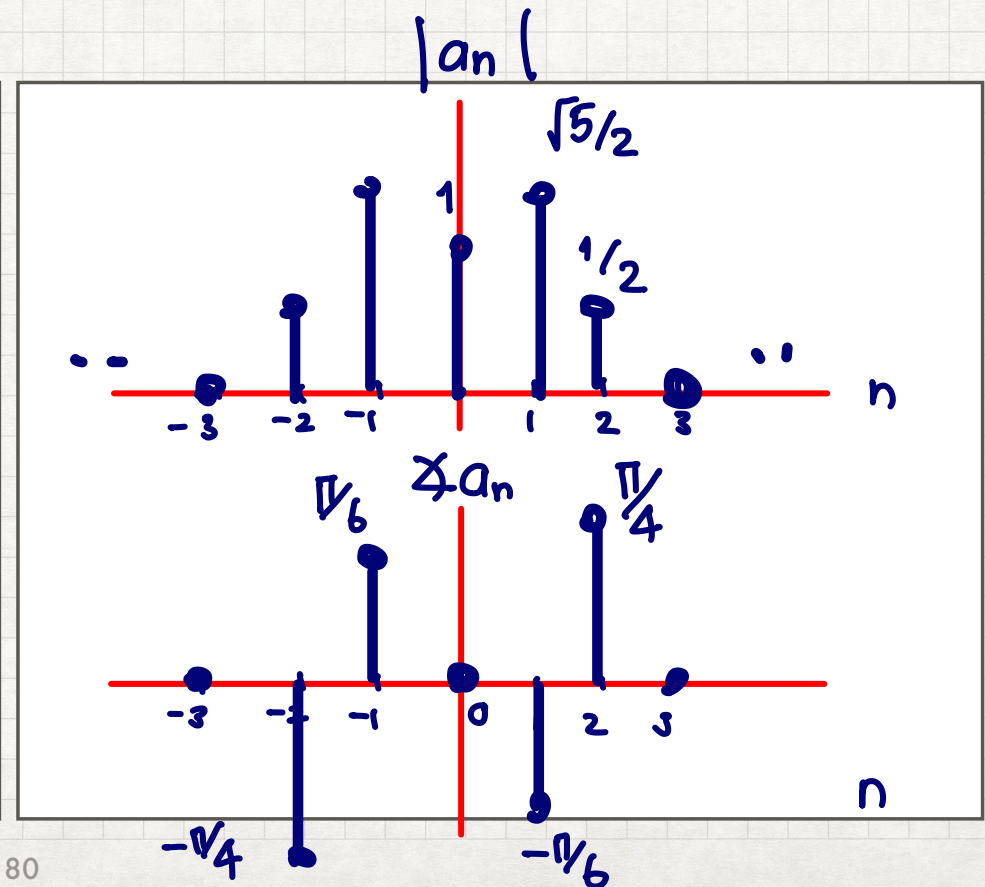
$$a_1 = 1 - \frac{j}{2}; |a_1| = \sqrt{1 + \frac{1}{4}} = \frac{\sqrt{5}}{2}; \angle a_1 = \tan^{-1}(-\frac{1/2}{1}) = -26.57^\circ$$

$$a_{-1} = 1 + \frac{j}{2}; |a_{-1}| = \frac{\sqrt{5}}{2}; \angle a_{-1} = \frac{\pi}{6}$$

$$a_2 = \frac{1}{2}e^{j\frac{\pi}{4}} = \frac{\sqrt{2}}{4} + j\frac{\sqrt{2}}{4}; |a_2| = \frac{1}{2}; \angle a_2 = \frac{\pi}{4}$$

$$a_{-2} = \frac{1}{2}e^{-j\frac{\pi}{4}} = \frac{\sqrt{2}}{4} - j\frac{\sqrt{2}}{4}; |a_{-2}| = \frac{1}{2}; \angle a_{-2} = -\frac{\pi}{4}$$

80



$\omega n \rightarrow 14:45$

DEMO

- Ref.: Wikipedia commons



$$1) \quad 3 + 2\sin(2\omega_0 t) + \cos(\omega_0 t)$$

$\uparrow$
 a_0

$$2 \cdot \frac{1}{2j} (e^{2j\omega_0 t} - e^{-2j\omega_0 t})$$

$$a_2 = -j; |a_2| = 1, \angle a_2 = -\frac{\pi}{2}$$

$$a_{-2} = j, |a_{-2}| = 1, \angle a_{-2} = \frac{\pi}{2}$$

$$\cancel{a_2} = -\cancel{a_{-2}} \quad \text{not allowed}$$

$$\frac{1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t})$$

$$a_1 = a_{-1} = \frac{1}{2}$$

$$2) \quad \int_{-\infty}^{\infty} x(\tau) \delta(\tau - t) d\tau = x(t)$$

$$= x(t) * \delta(t) = x(t) \quad \checkmark$$

$\delta(t) \rightarrow \boxed{\Delta} \rightarrow \Delta$

$$\int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt = x(t_0) \quad \# \text{ sifting properties } \quad \checkmark$$

$$x(t) + s(t) = \int x(\tau) \delta(\tau - t) d\tau$$

$$= x(t - \tau) \delta(\tau)$$

$$\delta(-\tau + t)$$

$$\delta(\tau - t)$$

$$\frac{1}{|-1|} \delta(\tau - t)$$

$x(t)$

